

AMAN CHAUDHARY

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SUMMARY

M.Tech graduate in CS (AI & Robotics) building and evaluating LLM systems. Built a long-term memory layer for LLM agents that matched the best baseline at 97% on held-out tests while holding 88% fewer memories, plus RAG pipelines with hallucination guards, and shipped ML APIs and full-stack LLM-powered apps with FastAPI, Next.js, PostgreSQL and Docker.

TECHNICAL SKILLS

Languages	Python, C++, JavaScript, TypeScript, SQL
AI / LLM	LLMs, RAG, Agent Memory, Prompt Engineering, Agentic Workflows, LLM Evaluation, LLM-as-Judge, Vector Search, Embeddings, ChromaDB, FastEmbed, Groq (Llama, gpt-oss), Hugging Face Transformers, DistilBERT, NLP
ML	PyTorch, TensorFlow, Scikit-Learn, XGBoost, OpenCV, Pandas, NumPy, Feature Engineering, Model Evaluation
Web & APIs	FastAPI, Flask, REST APIs, React, Node.js, Express.js, PostgreSQL, SQLite, Redis, Streamlit
Cloud & Infra	Docker, AWS, Git, GitHub Actions CI/CD, pytest, Linux, Render, Vercel, Model Monitoring

EDUCATION

Integrated B.Tech-M.Tech in Computer Science Engineering (AI & Robotics)	2021 – June 2026
Gautam Buddha University, Greater Noida · CGPA: 8.45 / 10 · Top 5% of batch	

PUBLICATIONS & RESEARCH

- M.Tech Thesis: Hybrid ACO-GA Smart Contract Optimization** · Python · NumPy · Genetic Algorithm · Blockchain [GitHub](#)
- Accepted at NetCrypt 2026, JNU — a dynamic knowledge transfer framework compressing GA solutions into ACO pheromone trails for faster convergence.
 - Built a hybrid ACO-GA scheduler modeling gas price fluctuation, network congestion, and real-time vulnerability signals; 36% faster convergence, 11% fitness improvement, and 41% risk reduction over pure ACO.
 - Evaluated across 3 complexity levels (15/35/75 tasks, 12 resource nodes) under dynamic gas prices (1.0–2.1x), congestion (10–80%), and simulated security threats against ACO and random baselines.

PROJECTS

- lethe — Long-Term Memory for LLM Agents** · Python · FastAPI · ChromaDB · Groq [GitHub](#) | [Live Demo](#)
- Built a long-term memory layer for LLM agents: extracts durable facts from chats, keeps a small active set in ChromaDB (FastEmbed embeddings), archives the lowest-scoring memories and reloads them when a question needs them.
 - Built an eval harness with 4 baselines, held-out chats, an LLM judge and 3 repeated runs with 95% CIs; matched the best baseline at 97% (CI 93–100%) with 88% fewer memories (3 vs 25) and ~52% less text per answer than full history.
 - Ordered recalled memories chronologically, lifting accuracy on questions about changed facts from 38% to 100% (retrieval baseline).
 - Deployed on Render with a live UI showing what memory keeps, archives and recalls; 49 tests and Ruff in GitHub Actions CI.
- Real-Time NLP Sentiment Analyzer** · Python · Flask · Scikit-Learn · Docker [GitHub](#) | [Live Demo](#)
- Benchmarked TF-IDF + Logistic Regression (93.3% accuracy, 0.03ms/input) against a DistilBERT transformer (Hugging Face); TF-IDF proved 259x faster — optimal for sub-2ms production serving, with DistilBERT identified as the accuracy upgrade path (97.5% on binary classification).
 - Containerised with Docker; documented an OpenAPI 3.0 spec; tested end-to-end with Postman.
- Assessment Recommendation Engine (RAG)** · Python · FastAPI · Groq LLaMA · Custom RAG [GitHub](#) | [Live Demo](#)
- Built a retrieval-augmented recommendation system over an 84-item public assessment catalog (SHL) using FastAPI and Groq LLaMA; engineered a hallucination guard constraining LLM output to retrieved catalog items only.
 - Measured retrieval quality with a 50-query tiered eval set scored on Recall@5 and MRR, using the results to tune the ranking layer.
 - Deployed as a live REST API on Render with interactive OpenAPI (Swagger) docs and structured JSON responses for downstream integration.
- DevPulse — Async Standup & Blocker Tracker** · TypeScript · Next.js · PostgreSQL · Redis [GitHub](#) | [Live Demo](#)
- Integrated the Groq LLaMA API to generate structured daily standup summaries with prompt-engineered JSON output; JWT auth, PostgreSQL schema, Redis caching, and 17 Jest/Supertest tests running in GitHub Actions CI; deployed on Vercel and Render.

EXPERIENCE

- Cyber Security Intern — Amroha Police, Uttar Pradesh** [Link](#) June – July 2025
- Conducted OSINT investigations using public records databases across 10+ cybercrime cases to support active threat identification.
 - Identified 3+ web vulnerabilities (XSS, IDOR) using Burp Suite via bug bounty participation; competed in CTF challenges covering cryptography, network forensics, and binary exploitation.
 - Prepared technical documentation and incident reports for cybercrime cases, translating investigation findings into structured summaries for law enforcement records.

CERTIFICATIONS

Supervised Machine Learning — DeepLearning.AI & Stanford University (Andrew Ng) · Coursera Link	May 2026
Intermediate Machine Learning — Kaggle Link	June 2026
Blockchain and its Applications — NPTEL Link	May 2024